Global Research Proposal - BIT/Nestlé

AI-Driven Multimodal Accessibility Platform for Inclusive Workplace Communication

Background

In large multinational organizations like Nestlé, communication relies heavily on visual and audio content—ranging from presentations and scanned documents to video meetings. However, employees with visual or hearing impairments face barriers in accessing this information. An integrated AI-driven accessibility platform, leveraging Optical Character Recognition (OCR) and Speech-to-Text (STT), can ensure inclusivity by automatically extracting and transcribing content, while also adapting outputs to individual needs.

This research initiative will be split into three complementary subtopics, each addressing a critical AI challenge, with the final goal of delivering a unified accessibility platform integrated into corporate tools such as Outlook, Teams, and websites.

Topic 1: Deep Learning for OCR in Real-World Corporate Documents

Objective:
To design, evaluate, and fine-tune deep learning OCR models for extracting text from diverse corporate visual materials (e.g., slides, PDFs, scanned reports, screenshots).

Research Questions:

1. How can OCR models be optimized to handle noisy, low-resolution, or multilingual corporate documents?

2. What fine-tuning strategies best adapt OCR models to Nestlé-specific vocabulary (brands, technical terms)?

3. How can OCR outputs be structured for seamless integration with accessibility tools (screen readers)?

Methodology:

- Conduct literature review on deep learning-based OCR.
- Build a dataset of corporate-like documents (scans, screenshots, slides).
- Benchmark models such as TrOCR, LayoutLM, Donut, and Tesseract.
- Apply transfer learning and fine-tuning with domain-specific data.
- Evaluate with Character Error Rate (CER), Word Error Rate (WER), and accessibility compliance.

Expected Outcome:
A fine-tuned OCR pipeline capable of extracting accurate text from complex workplace visuals, ready for integration into Outlook, Teams, and web platforms.

Topic 2: Advanced Speech-to-Text Models for Noisy, Multilingual Meetings

Objective:
To develop robust AI-based STT systems that can accurately transcribe corporate meetings and video calls in real time, handling multiple speakers, background noise, and multilingual content.

Research Questions:

1. How do state-of-the-art transformer-based ASR models perform in noisy and multilingual corporate settings?
2. Which adaptation strategies (transfer learning, domain-specific fine-tuning) improve transcription accuracy for workplace conversations?

3. How can real-time transcription latency be minimized without sacrificing accuracy?

Methodology:

- Literature review on advanced ASR methods.
- Collect/curate datasets simulating corporate meetings with multilingual and accented speech.
- Evaluate ASR models such as Whisper, wav2vec 2.0, and Conformer.
- Experiment with noise reduction, speaker diarization, and domain adaptation.
- Test prototypes for real-time transcription in simulated Teams meetings.

Expected Outcome:
A real-time STT engine optimized for corporate communications, capable of delivering accessible transcripts for hearing-impaired employees.

Topic 3: AI-Driven Personalization and Accessibility Layer

Objective:
To design an intelligent accessibility interface that personalizes OCR and STT outputs, making them adaptive, explainable, and inclusive for employees with diverse accessibility needs.

Research Questions:

1. How can AI personalization improve accessibility for visually and hearing-impaired users?
2. What explainable AI methods (confidence indicators, error highlighting) increase trust in AI outputs?

3. How can reinforcement learning from user feedback continuously improve accessibility features?

Methodology:

- Literature review on personalization, explainable AI, and adaptive accessibility design.
- Prototype interfaces with customizable features (live captions, adjustable fonts/contrast, multi-language subtitles).
- Integrate confidence scoring and error feedback loops.
- Conduct user studies with accessibility tool users (screen reader users, hearing-impaired participants).
- Explore reinforcement learning approaches for adaptive improvement.

Expected Outcome:
An adaptive accessibility layer that transforms raw OCR and STT outputs into inclusive, user-friendly communication tools, ensuring compatibility with Outlook, Teams, and assistive technologies.

Final Integration

- **Topic 1 (OCR Engine):** Extracts text from images, scanned documents, and slides.
- **Topic 2 (STT Engine):** Transcribes spoken content from meetings and videos in real time.
- **Topic 3 (Accessibility Layer):** Adapts and personalizes the extracted content for inclusive use.

Together, these three research paths will produce a comprehensive AI-powered accessibility platform that fosters inclusivity, enhances communication, and supports Nestlé's goal of creating an accessible workplace for all employees.